Linux Commands

1. pgrep – Command explanation

Command: pgrep -a bash

Purpose: Finds processes whose name matches bash. The -a option displays the PID and full command line.

Observed result: Two Bash processes appeared, with PIDs 426 and 544.

2. sleep – Command explanation

Command: sleep 300 & ; pgrep -a sleep ; pkill -x sleep

Purpose: Starts a 300-second sleep process in the background, finds it, then terminates a process whose name exactly matches sleep. The -x option enforces an exact name match.

Observed result: The test sleep process had PID 595 and was reported as Terminated. Use pkill carefully because it can stop real applications.

3. sleep – Command explanation

Command: sleep 200 & ; jobs -l

Purpose: Starts a background sleep process and lists jobs belonging to the current shell. The -l option includes process IDs.

Observed result: Job [1], PID 617, appeared as Running. jobs is not a system-wide process list.

4. pidof – Command explanation

Command: pidof systemd

Purpose: Displays the process ID of a running program named systemd.

Observed result: Your output returned PID 497. PIDs vary between sessions.

5. curl – Command explanation

Command: curl -I https://example.com

Purpose: Sends an HTTP HEAD request and displays response headers without downloading the page body.

Observed result: The server returned HTTP/2 200, indicating a successful response.

6. curl – Command explanation

Command: curl -o example.html https://example.com

Purpose: Downloads the response and saves it as example.html in the current directory.

Observed result: The transfer summary showed 577 bytes received. Verify with ls -l example.html.

7. nslookup – Command explanation

Command: nslookup google.com

Purpose: Queries a DNS resolver for IP address information about a domain name.

Observed result: The first screenshot showed DNS server 10.255.255.254 and returned IPv4 142.250.183.174 and IPv6 2404:6800:4007:834::200e.

8. ss – Command explanation

Command: ss -s

Purpose: Displays a summary of network sockets, including TCP, UDP, RAW, and other socket types.

Observed result: Your output showed 211 total sockets. Counts change with network activity.

9. realpath – Command explanation

Command: realpath file1.txt

Purpose: Resolves a path and prints the absolute path of the file.

Observed result: Output: /home/Srini/file1.txt.

10. ln – Command explanation

Command: ln -s file1.txt shortcut.txt

Purpose: Creates a symbolic link named shortcut.txt pointing to file1.txt. It is a reference, not a separate copy.

Observed result: You created the link. Verify with ls -l shortcut.txt.

11. readlink – Command explanation

Command: readlink shortcut.txt

Purpose: Prints the target stored in a symbolic link.

Observed result: Output: file1.txt. This is a relative link target.

12. traceroute – Command explanation

Command: traceroute google.com

Purpose: Shows the network hops between your computer and the destination.

Observed result: Hops 1–3 were 172.31.224.1, 192.168.0.1, and 192.168.1.1. Later asterisks mean probes received no reply before timeout.